INCH POUND

MIL-DTL-32633
31 July 2019

DETAIL SPECIFICATION

CONNECTORS, FILTER, ELECTROMAGNETIC INTERFERENCE (EMI), GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for connectors that contain filters for the suppression of electromagnetic interference (EMI). Filter connectors furnished to the requirements of this specification contain capacitor arrays and ferrite beads as applicable for C, Pi, L-C, C-L, or T filter circuits.

1.2 Classification. Filter connectors mate with MIL-DTL-38999 connectors and are of the following series, class, temperature ranges, and contact styles, as specified.

1.2.1 Series. Filter connectors are series III, scoop-proof, triple start, self-locking, threaded coupling connectors, in accordance with MIL-DTL-38999.

1.2.2 Class. Filter connector class is in accordance with MIL-DTL-38999, series III.

1.2.3 Temperature ranges. The upper temperature is the maximum internal hotspot temperature resulting from any combination of electrical load and ambient conditions. The filter connector temperature range is in accordance with MIL-DTL-38999, as defined for series III connector and finish, but not to exceed +125 °C.

Comments, suggestions, or questions on this document should be addressed to: Naval Air Systems Command (Commander, Naval Air Warfare Center Aircraft Division Lakehurst, Code 4.1.2.2, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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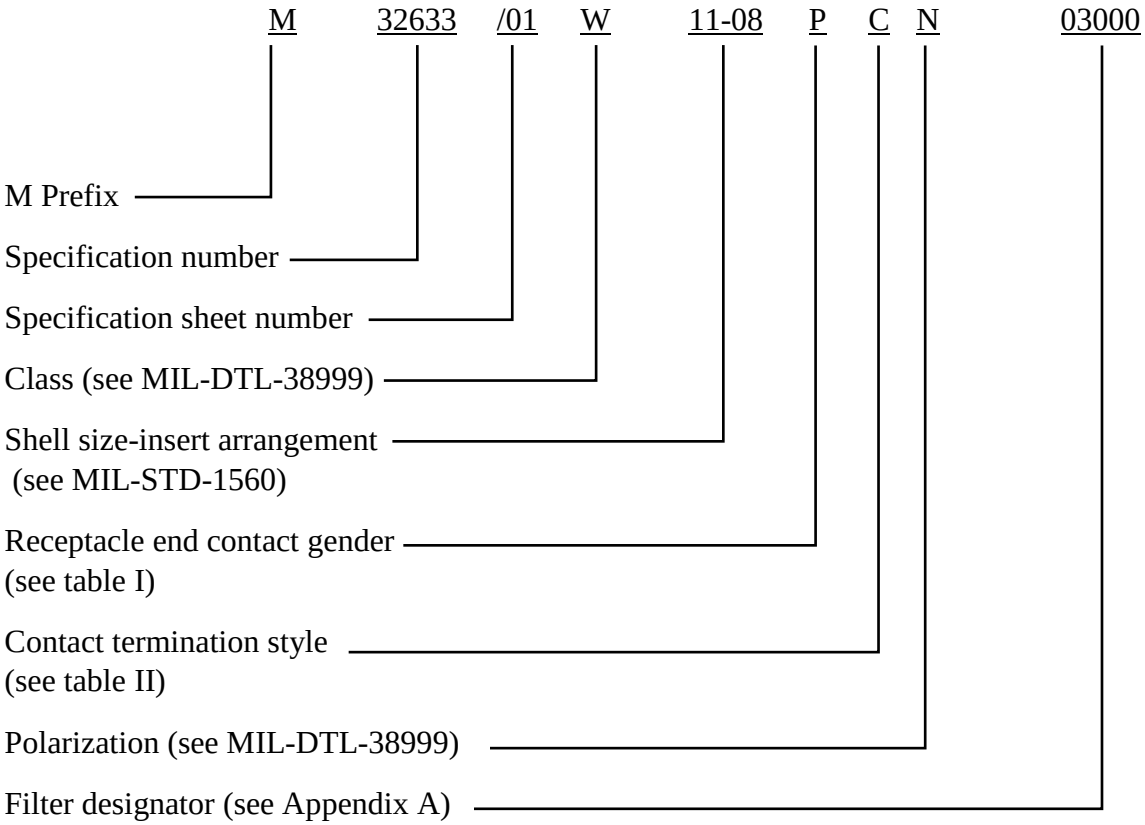
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1.2.4 Contact styles. Contact styles are as follows:

P – Pin - 500 cycle contact
S – Socket - 500 cycle contact
C – Crimp

1.3 Part or Identifying number (PIN). The PIN to be used for connectors acquired to this specification is created as follows:

Part or Identifying Number (PIN) example:



1.3.1 Filter designator. The filter type, filter characteristics, ceramic material and resistance to shell will vary based on the filter connector application. Assignment of a filter designator number is defined in Appendix A, table I.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-5606	-	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance
MIL-PRF-7808	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-DTL-38999	-	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, or Breech Coupling), Environment Resistant with Crimp Removable Contacts or Hermetically Sealed with Fixed, Solderable Contacts, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Electronic and Electrical Component Parts
MIL-STD-202-108	-	Method 108, Life (at Elevated Ambient Temperature)
MIL-STD-202-210	-	Method 210, Resistance to Soldering Heat
MIL-STD-202-305	-	Method 305, Capacitance
MIL-STD-220	-	Method of Insertion Loss Measurement
MIL-STD-790	-	Established Reliability and High Reliability Qualified Products List (QPL) Systems for Electrical, Electronic, and Fiber Optic Parts Specifications
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1285	-	Marking of Electrical and Electronic Parts
MIL-STD-1560	-	Insert Arrangements for MIL-DTL-38999, MIL-DTL-27599, and SAE AS29600 Series A Electrical Circular Connectors

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)

ECIA EIA-364-06	-	Contact Resistance Test Procedure for Electrical Connectors
ECIA EIA-364-20	-	Dielectric Withstanding Voltage Test Procedure For Electrical Connectors, Sockets, and Coaxial Contacts
ECIA EIA-364-21	-	Insulation Resistance Test Procedure For Electrical Connectors, Sockets, and Coaxial Contacts
ECIA EIA-364-28	-	Vibration Test Procedure for Electrical Connectors and Sockets
ECIA EIA-364-32	-	Thermal Shock (Temperature Cycling) Test Procedure for Electrical Connectors and Sockets
ECIA EIA-364-79	-	Insert Bond Strength Test Procedure for Electrical Connectors
ECIA EIA-364-114	-	Coupling and Uncoupling Test Procedure for Electrical Connectors, Sockets and Applicable Accessories

(Copies of these documents are available online at http://www.ecianow.org.)

RADIO TECHNICAL COMMISSION FOR AERONAUTICS (RTCA)

DO-160	-	Environmental Conditions and Test Procedures for Airborne Electrical/Electronic Equipment and Instruments

(Copies of this document are available from http://www.rtca.org.)

SAE INTERNATIONAL

SAE AS22520	-	Crimping Tools, Wire Termination, General Specification for
SAE EIA-557	-	Statistical Process Control Systems

(Copies of these documents are available from http://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be specified herein and in accordance with the applicable specification sheets. In the event of a conflict between the requirements of this specification and the specification sheet, the latter shall govern.

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3.2 Qualification. Filter connectors furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable Qualified Products List (QPL) before contract award (see 4.3, 4.4, and 6.3).

3.2.1 Qualified Products List (QPL) system. The manufacturer shall establish and maintain a quality system that allows its parts that are covered by this specification to be listed on the QPL. Requirements for this system are specified in MIL-STD-790. In addition, the manufacturer shall establish a Statistical Process Control (SPC) system that meets the requirements specified in SAE EIA557.

3.3 Mating connector. The military part number of the connector to which the filter connector is intended to mate is specified in applicable specification sheet (see 3.1).

3.4 Shell style. The shell style of the filter connector, if applicable, shall be as specified herein and in accordance with the specification sheet. A filter connector can be a hermetic or non-hermetic receptacle, which is a box mount, wall mount, through bulkhead mount (receptacle interface on each side of the mounting flange) or jam nut mount. The hermetic or non-hermetic filter connector can also be a plug or connector adapter (see 6.15).

3.5 Filter connector class. The filter connector class, if applicable, shall be the same as and meet the requirements of the MIL-DTL-38999 connector to which the filter connector is intended to mate. The part number of the mating connector shall be in accordance with the specification sheet (see 3.1).

3.6 Insert arrangement. The insert arrangement (see MIL-STD-1560) shall be the mirror image of the MIL-DTL-38999 connector to which the filter connector is intended to mate, thereby ensuring proper mating of the connectors.

3.7 Contact genders. Pin contacts used in the filter connector shall mate with socket contacts of the mating MIL-DTL-38999 connector, and socket contacts used in the filter connector shall mate with pin contacts in the mating MIL-DTL-38999 connector. See contact genders in table I.

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TABLE I. Contact genders.

Symbol	Contact gender
P	Pin contact insert
S	Socket contact insert
A	Pin plug adapter 1/
B	Socket plug adapter 1/

1/ These symbols are only applicable to the connector adapter (see 6.15).

3.8 Contact termination. The contact termination shall be in accordance with table II and the specification sheet.

TABLE II. Contact termination styles.

Symbol	Contact termination style
C	Crimp
S	Solder cup
P	PC tail

3.9 Shell polarization. The shell polarization, if applicable, shall be the same as the polarization of the MIL-DTL-38999 connector to which the filter connector is intended to mate (see the specification sheet).

3.10 Voltage rating. The working voltage rating at sea level of the filter connector shall be 200 VDC or 115 V_{RMS} (400 Hz) or as specified in the specification sheet.

3.11 Filter type. The filter type shall be as shown on figure 1 and meet the requirements of table III and the specification sheet.

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TABLE III. Filter types.

Filter type	Application
C	High source and high load impedance
Pi	Unknown source or medium source and load impedance
C-L	High source and low load impedance 1/
L-C	Low source and high load impedance 2/
T	Low source and low load impedance

1/ The inductive element shall be positioned toward the terminating face of the connector.
2/ The inductive element shall be positioned toward the mating face of the connector.

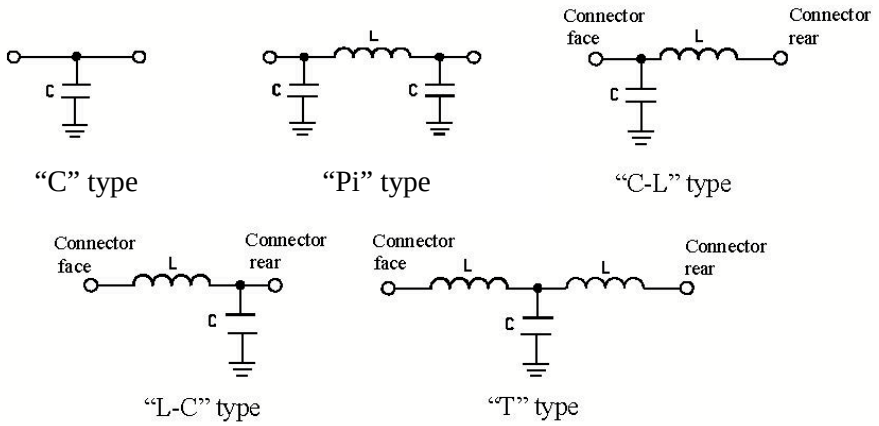


FIGURE 1. Filter types.

3.12 Filter characteristics. The filter characteristics shall be as specified in tables IV, V, and VI, as well as meet the requirements of the specification sheet.

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TABLE IV. "C" type filter electrical characteristics.

Characteristic	Capacitance (in picofarads [pF])		Minimum insertion loss (in decibels [dB])			
	Min	Max	Test frequencies			
			1 MHz	10 MHz	100 MHz	1000 MHz
A	19000	28000	6	24	41	50
B	16000	22500	5	23	39	49
C	9000	16500	3	16	35	46
D	4000	6000	-	8	28	41
E	1650	2500	-	4	21	34
F	400	650	-	-	10	23
G	200	300	-	-	5	17

TABLE V. "Pi" type filter electrical characteristics.

Characteristic	Capacitance (pF)		Minimum insertion loss (dB)			
	Min	Max	Test frequencies			
			1 MHz	10 MHz	100 MHz	1000 MHz
H	38000	56000	10	40	62	66
I	32000	45000	8	35	60	62
J	18000	33000	5	25	57	60
K	8000	12000	1	14	50	58
L	3300	5000	-	8	40	52
M	800	1300	-	2	15	32
N	400	600	-	0.8	13	22

3.13 Contact identification. Each contact shall be identified as ground, feed-through, or filtered, as applicable (see specification sheet). Filtered contacts shall be identified by the applicable filter type and filter characteristic (see 3.11 and 3.12).

3.14 Materials. The materials shall be as specified herein. However, when a definite material is not specified, a material shall be used that enables the filter connectors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.

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3.14.1 Pure tin. The use of pure tin as an underplate or final finish is prohibited both internally and externally on all filter connector components. Tin content of filter connector components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.4).

3.14.2 Connector components. Individual components within the filter connector shall meet the component requirements of the connector to which the filter connector is intended to mate (see the specification sheet).

3.14.2.1 Contact material and plating. The contact material and plating shall be the same as and meet the requirements of the contact material and plating in the MIL-DTL-38999 connector to which the filter connector is intended to mate (see specification sheet).

3.14.2.2 Shell material and finish. The shell material and finish shall be the same as the shell material and finish of the MIL-DTL-38999) connector to which the filter connector is intended to mate (see specification sheet).

TABLE VI. "C-L", "L-C", and "T" type filter electrical characteristics.

Characteristic	Capacitance (pF)		Minimum insertion loss (dB)			
	Min	Max	Test frequencies			
			1 MHz	10 MHz	100 MHz	1000 MHz
O	19000	28000	6	25	42	52
P	16000	22500	6	24	41	51
Q	9000	16500	3	17	38	48
R	4000	6000	-	9	30	43
S	1650	2500	-	6	23	36
T	400	650	-	-	12	25
U	200	300	-	-	6	19

3.14.3 Ceramic material. The material used in the manufacture of capacitor arrays shall have the characteristics specified in table VII.

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TABLE VII. Capacitance change with reference to +25 °C.

Ceramic capacitor characteristic 1/	Steps A through D of table XV Bias = 0 volts	Steps E through G of table XV Bias = rated voltage
BP	0 ±30 ppm/°C	0 ±30 ppm/°C
BX	±15 percent	+15, -25 percent
C0G	0 ±30 ppm/°C	N/A
X7R	±15 percent	N/A

1/ See 6.11 for characteristic definitions.

3.15 Interface and physical dimensions. The filter connectors shall meet the interface dimensions of the MIL-DTL-38999 connector to which it is intended to mate (see specification sheet).

3.16 Mounting. Mounting requirements (front or rear), if applicable, shall be in accordance with the specification sheet.

3.17 Interchangeability. The filter connectos shall meet the interchangeability requirements of the MIL-DTL-38999 connector to which it is intended to mate (see specification sheet).

3.18 Visual inspection. The filter connectors shall be inspected for anomalies as specified in 4.6.1.

3.19 Electrical performance.

3.19.1 Dielectric withstanding voltage (DWV). When filter connectors are tested as specified in 4.6.2, there shall be no breakdown, flashover, or impairment of any characteristics sufficient to cause failure of the filter connector.

3.19.2 Insulation resistance (IR). Unless otherwise required in the specification sheet, when filter connectors are tested as specified in 4.6.3, the insulation resistance shall meet the following requirements:

a. Individual contacts:
   (1) At +25 °C: Not less than 5 gigaohms.
   (2) At +125 °C: Not less than 1 gigaohm.

b. Contacts connected in parallel (see 6.8):
   (1) At +25 °C: Not less than 5 gigaohms divided by the number of contacts connected in parallel (see 6.8).
   (2) At +125 °C: Not less than 1 gigaohm divided by the number of contacts connected in parallel (see 6.8).

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Note: The specified IR limits are set low to support automatic contact testers. Higher limits would result in excessively long test times in multi-pin filter connectors. Actual IR readings may be 1 or more orders of magnitude greater than minimum limits specified. Insulation resistance (+25 °C) may be combined with DWV by measuring the insulation resistance at the DWV level. If combined with DWV, there shall be no evidence of breakdown or flashover.

3.19.3 Capacitance to ground. When filter connectors are tested as specified in 4.6.4, the capacitance to ground shall be as specified herein and in accordance with the specification sheet (see 3.1).

3.19.4 Crosstalk capacitance. Unless otherwise required in the specification sheet, when the filter connectors are tested as specified in 4.6.5, the crosstalk capacitance to any adjacent or nonadjacent contact shall be less than 25 pF for capacitor characteristics BX and X7R or less than 10 pF for capacitor characteristics BP or C0G.

3.19.5 Dissipation factor. Unless otherwise required in the specification sheet, when filter connectors are tested as specified in 4.6.6, the dissipation factor shall not be greater than 2.5 percent for BX or X7R capacitor characteristic, 0.15 percent for BP or C0G capacitor characteristic.

3.19.6 Insertion loss. When filter connectors are tested as specified in 4.6.9, the insertion loss shall not be less than specified herein and in the specification sheet.

3.20 Thermal shock. When tested as specified in 4.6.7, filter connectors shall show no evidence of cracking or fracture. In addition, filter connectors shall meet the electrical performance requirements (sea level, +25 °C, see 3.19).

3.21 Voltage conditioning. When tested as specified in 4.6.8 (see 6.8), filter connectors shall meet the electrical performance (sea level, +25 °C, see 3.19) requirements. The insulation resistance (+125 °C) shall be as specified in 3.19.2 for contacts connected in parallel.

3.22 Crosstalk insertion loss. Unless otherwise required in the specification sheet (see 3.1), when filter connectors are tested as specified in 4.6.10, the crosstalk insertion loss to any adjacent or nonadjacent contact shall be 80dB, minimum at 100 MHz.

3.23 Dielectric characteristic. When tested as specified in 4.6.11, the capacitance change over the specified temperature range shall not exceed the limits specified in table VII. The capacitance value measured during testing (see 4.6.11) at step C of table XV shall be considered as the reference point.

3.24 Contact retention. When tested as specified in 4.6.12, filter connector inserts shall retain their contacts. The axial displacement of contacts shall not exceed 0.012 inch while under the test load, unless otherwise specified in the applicable specification sheet. In addition, filter connectors shall meet the electrical performance (see 3.19) and visual inspection (see 3.18) requirements. The insulation resistance (+125 °C) shall be as specified in 3.19.2 for contacts connected in parallel and DWV shall be

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as specified at sea level (see 3.19.1).

3.25 Contact engagement and separating force (socket contacts only in filter connector). When tested as specified in 4.6.13, filter connectors shall meet the requirements of the connector to which it is intended to mate (see specification sheet and 3.1).

3.26 Voltage spike (when required in the specification sheet). When tested as specified in 4.6.14, filter connectors shall show no evidence of flashover, degradation, or breakdown. In addition, filter connectors shall meet the electrical performance requirements (sea level, +25 °C, see 3.19).

3.27 Lightning strike (when specified in the specification sheet). When tested as specified in 4.6.15, filter connectors shall show no evidence of flashover, degradation, or breakdown. In addition, filter connectors shall meet the electrical performance requirements (sea level, +25 °C, see 3.19) and visual inspection requirements (see 3.18).

3.28 Radio frequency (RF) current. When required in the specification sheet, the connector shall be tested as specified in 4.6.16 and shall show no degradation. In addition, filter connectors shall meet the electrical performance requirements (sea level, +25 °C, see 3.19).

3.29 Contact resistance. When tested as specified in 4.6.17 using the test current and wire size specified in table VIII, contacts in the mated condition shall meet the contact resistance requirements in table VIII.

TABLE VIII. Contact resistance.

Contact size	Wire size	Test current	Maximum millivolt drop (millivolts)	
			Initial	After voltage conditioning 2/
22	.020 1/	3.0	70	75
22	22	5.0	73	88
20	.030 1/	5.0	65	70
20	20	7.5	55	66
16	16	13.0	49	59

1/ Printed circuit board contacts.
2/ See 3.21 voltage conditioning.

3.30 Ground contact resistance (when specified herein or in the specification sheet, see 3.1). When filter connectors are tested as specified in 4.6.18, the resistance shall be no more than 15.0 milliohms for Pi filter and 30 milliohms for a C filter.

3.31 Water immersion. When tested as specified in 4.6.19, filter connectors shall show no

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evidence of air leakage as demonstrated by repetitive bubbling. In addition, filter connectors shall meet the electrical performance requirements (see 3.19) at sea level and +25 °C, except crosstalk capacitance (see 3.19.4).

When vacuum dried as specified in 4.6.19f, filter connectors shall meet the following requirements:

a. Dielectric withstanding voltage (at sea level): As specified in 3.19.1.
b. Insulation resistance (+25 °C), as specified in 3.19.2.

3.32 Insert retention. When tested as specified in 4.6.20, the inserts of filter connectors shall not be dislocated from their original positions when a pressure of 100 ±5 psi with a minimum force of 25 pounds-force is applied to the insert.

3.33 Fluid immersion (when required herein or in the specification sheet, see 3.1). When tested as specified in 4.6.21, filter connectors shall meet the coupling torque requirements of the MIL-DTL-38999 connector to which the filter connector is intended to mate. In addition, filter connectors shall meet the following requirements:

a. Capacitance to ground: As specified in 3.19.3.
b. Dissipation factor: As specified in 3.19.5.
c. Visual inspection: As specified in 3.18. In addition, filter connectors shall be visually examined with the unaided eye for evidence of degradation such as:
   (1) Excessive swelling. Swelling is permitted provided that the connector coupling torque remains within requirement limits.
   (2) Cracking or pitting of any resilient material.
   (3) Loss of seal or adhesive bonding between two bonded surfaces.
   (4) Softening of any material.

3.34 Shock. When tested as specified in 4.6.22 (see 6.9), filter connectors shall show no signs of damage or breakage, nor exhibit any discontinuities greater than 1 µsec. In addition, filter connectors shall meet the electrical performance requirements (see 3.19), at sea level and +25 °C.

3.35 Random vibration. When tested as specified in 4.6.23 (see 6.9), filter connectors shall not crack or break, nor exhibit any discontinuities greater than 1 µsec, or loosening of parts. Receptacles shall retain plugs in full engagement during vibration. In addition, filter connectors shall meet the electrical performance requirements (see 3.19), at sea level and +25 °C.

3.36 Humidity. When tested as specified in 4.6.24, filter connectors shall meet the requirements of the MIL-DTL-38999 connector to which it is intended to mate (see specification sheet). In addition, filter connectors shall meet the electrical performance requirements (see 3.19), at sea level and +25 °C.

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3.37 Resistance to soldering heat (when specified herein or in the specification sheet). When tested as specified in 4.6.25, filter connectors with solderable, non-removable contacts shall show no evidence of internal or external damage. In addition, filter connectors shall meet the electrical performance requirements (see 3.19), at sea level and +25 °C.

3.38 Salt spray (corrosion). When tested as specified in 4.6.26, filter connectors shall meet the requirements of the MIL-DTL-38999 connector to which it is intended to mate (see the specification sheet).

3.39 Life. When tested as specified in 4.6.27, filter connectors shall meet the electrical performance requirements (see 3.19), at sea level and +25 °C.

3.40 Coupling and uncoupling torque. When tested as specified in 4.6.28, filter connectors shall meet the coupling torque requirements of table IX for mating and un-mating and coupling and uncoupling for the MIL-DTL-38999 connector, protective covers, and stowage receptacles to which the filter connector is intended to mate (see specification sheet). In addition, filter connectors shall meet the electrical performance requirements (see 3.19), at sea level and +25 °C.

TABLE IX. Coupling and uncoupling torque.

Shell size	Maximum engagement and disengagement		Minimum disengagement	
	Pound inch	Newton meters	Pound inch	Newton meters
9	8	.9	2	.2
11	12	1.4	2	.2
13	16	1.8	2	.2
15	20	2.3	3	.3
17	24	2.7	3	.3
19	28	3.2	3	.3
21	32	3.6	5	.6
23	36	4.1	5	.6
25	40	4.6	5	.6

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3.41 Dielectric voltage breakdown. When measured as specified in 4.6.29, $X - 3\sigma$ (the mean - 3 standard deviations) shall be greater than 250 percent of the DC rated voltage of the filter connector.

3.42 Marking.

3.42.1 Filter connector marking. Filter connectors shall be marked in accordance with MIL-STD-1285, method I with the following information: JAN marking (see 3.42.3), part or identifying number (PIN), date code and lot symbol (LDC), and the manufacturer's commercial and government entity (CAGE) number. Metal stamping shall be accomplished before plating. Additionally, the outside surface of the filter connector shell shall be marked with the words "FILTER CONTACTS" in a contrasting color to the basic shell plating, in at least two places that are readily visible when servicing the mounted filter connector.

3.42.2 Contact location identification. Contact locations shall be identified as specified in MIL-DTL-38999 and MIL-STD-1560. All positions shall be identified on the front and rear faces of the insert, except where space limitations make this impracticable. Location of contact identifying characters shall be in close proximity to the holes, but need not be placed exactly where indicated in MIL-DTL-38999 and MIL-STD-1560. The preferred color of the contact identifying character is white. When the background is a color against which white is difficult to distinguish, a color will be chosen for which the identifying character can be easily distinguished. Underscoring of lower case cavity identifiers shall be in accordance with MIL-STD-1560, but is optional for insert arrangements for which no underscoring is indicated, provided that the insert pattern can be easily determined and legibility is not impaired.

3.42.3 JAN and J marking. The United States Government has adopted and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Items furnished under contracts or orders that either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets, the manufacturer shall completely remove the military part number and the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark "JAN" and Registration Number 2,577,735 for the certification mark "J".

3.43 Recycled, recovered, or environmentally preferable or biobased materials. Recycled, recovered, or environmentally preferable materials, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance

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requirements, and promotes economically advantageous life cycle costs.

3.44 Workmanship. Filter connectors shall be processed in such a manner as to be uniform in quality and shall be free from cold soldering, corrosion, pits, dents, cracks, rough sharp edges, misalignments, and other defects that will affect life, serviceability, or appearance.

3.45 Fungus resistance. Materials used in the construction of these connectors shall be fungus inert (see 4.6.30).

3.46 Insert bond strength. When tested as specified in 4.6.31, after 30 seconds, there shall be no visible separation or cracks at the bond joint while a tensile load of 100 psi is applied to the bond joint. The load shall be applied at a rate of 1 pound per second.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Conformance inspection (see 4.5).

4.2 Inspection conditions. Unless otherwise specified herein and in the applicable specification sheet, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202, except the relative humidity shall not exceed 75 percent. The accuracy of all test voltage measurements shall be within ±2.0 percent of the specified voltage and all test temperatures above +25 °C shall have a tolerance of +4 °C, -0 °C.

4.2.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment of a calibration system to control the accuracy of the measuring and test equipment (i.e. industry standard, military standard) shall be required.

4.2.1.1 QPL system. The manufacturer shall establish and maintain a system to meet the requirements of MIL-STD-790 and the qualification requirements herein (see 3.2.1). Evidence of such compliance shall be verified by the qualifying activity as a prerequisite for qualification and retention of qualification. Assembly plants shall be approved by the qualifying activity and listed on the applicable QPL. The qualified connector manufacturer shall certify that the assembly plant is approved for the distribution of the manufacturer's parts. Assembly plants shall use only piece parts (i.e. subcomponents such as connector shells and inserts) produced by the qualified manufacturer. No incoming inspection other than visual examination is required of certified piece parts obtained from the qualified manufacturer, except when there is cause for rejection. Assemblies produced at the assembly plant shall be subjected to inspection to ensure that the assembly process conforms with that established at the qualified manufacturing plant. Quality control requirements, including Government inspection surveillance, shall be the same as required for the qualified connector manufacturer.

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4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests in table X. The qualifying activity shall perform the Group 3 tests in table X. Unless otherwise specified in table X, sequential testing is required for each group. The qualifying activity shall perform or witness the particular tests as specified in table X, and the supplier shall perform the remaining tests. See 6.3.1 for the qualification inspection process.

4.3.1 Sample size. The complexity of filter connectors will vary depending upon the application requirements for the connector. The samples to be submitted for qualification shall be representative of that complexity. All samples shall be made by normal production means. The minimum number of samples for each test group shall be 2 connectors. The number of samples shall be 4, if size 16 contacts are to be qualified. The samples shall have a minimum of 2 ground contacts, one placed in the outer row of contacts and one in the innermost row. If qualification is requested for a variety of filter capacitances, the samples shall contain the most standard Pi type filter, D, and a minimum of 2 additional Pi filters, C, E, or F (see table IV, V, and VI). The shell sizes shall be size 17 and 25 with size 22 contacts and insert arrangements 17-35 and 25-35 (see MIL-STD-1560). If qualification is requested for pin and socket contacts, then one sample of each shell size shall be fitted with socket contacts. If qualification is requested for size 16 contacts, then samples of the same shell sizes noted above shall be also supplied again with pin and socket contacts.

In addition to the connector samples, two insert assemblies (bonded pin insert units) representing the insert patterns being qualified manufactured by normal production process shall also be provided. The purpose is to verify that the adhesive used to bond the interfacial to the dielectric pin insert is suitable for the intended purpose. If a socket insert with grommet is used in the connector assembly, they also shall be provided.

4.3.2 Qualification sample selection of limited product. If qualification is requested for a limited production, 2 samples shall be submitted for each test group that have been manufactured by normal production means in the shell size and filter configuration for which qualification is requested. Qualification shall be limited to the connector styles submitted for qualification.

4.3.3 Preparation of initial qualification samples. If crimp removable contacts are used, wiring of crimp-type contacts shall be accomplished using tools conforming to SAE AS22520, class I (see 6.10).

4.3.4 Failures. Any failure during the qualification inspections (see table X) is cause for refusal to grant qualification. A failure in the mating plug does not constitute a failure of the filter connector qualification test. The failed mating plug may be replaced.

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TABLE X. Initial qualification inspections.

Inspection	Requirement paragraph	Test paragraph	Test group		
			1	2	3
Visual inspection	3.15, 3.18, 3.42, 3.44	4.6.1	X	X	X
Dielectric withstanding voltage (sea level) 1/	3.19.1	4.6.2.1	X	X	X
Insulation resistance (+25 °C) 1/	3.19.2	4.6.3.1	X	X	X
Capacitance to ground 2/	3.19.3	4.6.4	X	X	X
Crosstalk capacitance	3.19.4	4.6.5	X	X	X
Dissipation factor	3.19.5	4.6.6	X	X	X
Insertion loss (+25 °C)	3.19.6	4.6.9.1	X	X	X
Crosstalk insertion loss	3.22	4.6.10	X	X	
Dielectric characteristic	3.23	4.6.11	X	X	
Contact retention	3.24	4.6.12	X	X	
Coupling and uncoupling torque	3.40	4.6.28	X	X	X
Contact engagement and separating force (socket contacts only)	3.25	4.6.13	X	X	
Voltage spike (when specified, see 3.1)	3.26	4.6.14	X		
Lightning strike (when specified, see 3.1)	3.27	4.6.15	X		
Radio frequency current	3.28	4.6.16	X		
Capacitance to ground	3.19.3	4.6.4	X		
Dielectric withstanding voltage (altitude)	3.19.1	4.6.2.2	X		
Insulation resistance (+125°C)	3.19.2	4.6.3.2		X	
Insertion loss (+125°C)	3.19.6	4.6.9.2		X	
Contact resistance	3.29	4.6.17	X	X	
Ground contact resistance (when specified, see 3.1)	3.30	4.6.18	X	X	
Thermal shock	3.20	4.6.7	X	X	
Water immersion	3.31	4.6.19		X	
Humidity	3.36	4.6.24	X		

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TABLE X. Initial qualification inspections - Continued.

Inspection	Requirement paragraph	Test paragraph	Test group		
			1	2	3
Life	3.39	4.6.27			X
Random vibration	3.35	4.6.23	X		
Shock	3.34	4.6.22	X		
Insert retention	3.32	4.6.20	X		
Contact retention	3.24	4.6.12	X		X
Salt spray	3.38	4.6.26		X	
Contact resistance	3.29	4.6.17	X	X	
Ground contact resistance (when specified, see 3.1)	3.30	4.6.18	X	X	
Coupling and uncoupling torque	3.40	4.6.28	X	X	X
Contact engagement and separating force (socket contacts only)	3.25	4.6.13	X	X	
Fluid immersion, lubricating oil (when specified, see 3.1) 3/	3.33	4.6.21	X		
Fluid immersion, hydraulic fluid (when specified, see 3.1) 3/	3.33	4.6.21		X	
Resistance to soldering heat	3.37	4.6.25	X		
Insertion loss (-55°C)	3.19.6	4.6.9.3	X	X	X
Fungus resistance	3.45	4.6.30		X	
Insert bond strength	3.46	4.6.31	X	X	
Dielectric voltage breakdown	3.41	4.6.29	X		
Visual inspection	3.18, 3.42, 3.44	4.6.1	X	X	X

1/ Dielectric withstanding voltage and insulation resistance (+25 °C) may be tested simultaneously.
2/ Capacitance should be measured prior to the application of voltage. The application of voltage will temporarily polarize the dielectric and depress the capacitance.
3/ With approval of the qualifying activity, the fluid immersion test may be omitted if the manufacturer can certify that the materials used in the filter connector are the same as qualified during qualification testing for the connector to which the filter connector is intended to mate. Deletion of the tests does not relieve the manufacturer from meeting this test requirement.

4.4 Retention of qualification. Retention of qualification inspection shall occur every 36 months after the initial qualification date. The qualifying activity shall perform or witness the tests as specified in table XI. Unless otherwise specified in the table XI, sequential testing is required. The qualifying activity may shift the reporting date to accommodate testing schedules, but no later than 18 months from the retention of qualification due date.

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TABLE XI. Retention of qualification inspections.

Inspection	Requirement paragraph	Test paragraph	Test group	
			1	2
Visual inspection	3.15, 3.18, 3.42, 3.44	4.6.1	X	X
Dielectric withstanding voltage (sea level) 1/	3.19.1	4.6.2.1	X	X
Insulation resistance (+25 °C) 1/	3.19.2	4.6.3.1	X	X
Capacitance to ground 2/	3.19.3	4.6.4	X	X
Crosstalk capacitance	3.19.4	4.6.5	X	X
Dissipation factor	3.19.5	4.6.6	X	X
Insertion loss (+25 °C)	3.19.6	4.6.9.1	X	X
Crosstalk insertion loss	3.22	4.6.10	X	X
Dielectric characteristic	3.23	4.6.11	X	
Contact retention	3.24	4.6.12	X	X
Contact engagement and separating force (socket contacts only)	3.25	4.6.13	X	X
Coupling and uncoupling torque	3.40	4.6.28	X	X
Voltage spike (when specified, see 3.1)	3.26	4.6.14	X	
Lightning strike (when specified, see 3.1)	3.27	4.6.15	X	
Radio frequency current	3.28	4.6.16	X	
Capacitance to ground	3.19.3	4.6.4	X	
Dielectric withstanding voltage (altitude)	3.19.1	4.6.2.2	X	
Insulation resistance (+125 °C)	3.19.2	4.6.3.2		X
Insertion loss (+125 °C)	3.19.6	4.6.9.2		X
Contact resistance	3.29	4.6.17	X	X
Ground contact resistance (when specified, see 3.1)	3.30	4.6.18	X	X
Thermal shock	3.20	4.6.7	X	X
Water immersion (when specified, see 3.1)	3.31	4.6.19	X	
Life (500 hours)	3.39	4.6.27		X
Random vibration	3.35	4.6.23	X	
Insert retention	3.32	4.6.20	X	
Contact retention	3.24	4.6.12	X	
Contact resistance	3.29	4.6.17	X	X
Ground contact resistance (when specified, see 3.1)	3.30	4.6.18	X	X
Coupling and uncoupling torque	3.40	4.6.28	X	X
Insertion loss (-55 °C)	3.19.6	4.6.9.3	X	X
Visual inspection	3.15, 3.18, 3.42, 3.44	4.6.1	X	X

1/ Dielectric withstanding voltage and insulation resistance (+25 °C) may be tested simultaneously.

2/ Capacitance should be measured prior to the application of voltage. The application of voltage will temporarily polarize the dielectric and depress the capacitance.

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4.4.1 Preparation of retention of qualification samples. If crimp removable contacts are used, wiring of crimp-type contacts shall be accomplished using tools conforming to SAE AS22520, class I (see 6.10).

4.4.2 Samples for retention of qualification. Retention of qualification samples shall consist of a minimum of 2 samples for each test group in the largest shell size, 25, for size 22 contacts and size 16 contacts. There should be socket and pin contacts as noted with a mix of ground, feed through and filter contact types as noted in 4.3.1.

4.4.3 Failures. Any failure during retention of qualification inspections (see table XI) is cause for failure of the filter connector qualification test. A failure in the mating plug does not constitute a failure of the filter connector retention of qualification test. A failed mating plug may be replaced.

4.5 Conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A and group B inspections (see tables XII and XIII).

4.5.1.1 Inspection lot. An inspection lot shall consist of all filter connectors of a single PIN, produced under the same conditions and offered for inspection at one time.

4.5.2 Group A inspection. Group A inspection shall consist of the inspections specified in table XII and shall be performed on 100 percent of the product supplied under this specification. Filter connectors failing the tests of group A shall be removed from the lot.

TABLE XII. Group A inspections.

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Subgroup 1			
Dielectric withstanding voltage (at sea level) 1/	3.19.1	4.6.2.1	100%
Insulation resistance (+25 °C) 1/	3.19.2	4.6.3.1	
Capacitance to ground 2/	3.19.3	4.6.4	
Dissipation factor	3.19.5	4.6.6	
Visual and mechanical inspection	3.15, 3.18, 3.42, 3.44	4.6.1	
Subgroup 2			
Crosstalk capacitance	3.19.4	4.6.5	1 filter connector (see 4.5.2)

1/ Dielectric withstanding voltage and insulation resistance (+25 °C) may be tested simultaneously.
2/ Capacitance should be measured prior to the application of voltage. The application of voltage will temporarily polarize the dielectric and depress the capacitance.

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4.5.2.1 Subgroup 1. The tests of subgroup 1 shall be performed on 100 percent of the product supplied under this specification. Filter connectors failing the tests of subgroup 1 shall be removed from the lot. Lots having more than 5 percent total rejects shall not be furnished on the contract or purchase order.

4.5.2.2 Subgroup 2. The tests of subgroup 2 shall be performed on a sample filter connector that has passed the tests of subgroup 1. The sample shall be randomly selected. No failures are allowed.

4.5.3 Group B inspection. Group B inspection shall consist of the inspections specified in table XIII in the order shown. The group B inspection shall be performed on sample units selected at random from each inspection lot that has passed the group A inspection.

4.5.3.1 Subgroup 1. Samples subjected to the tests of subgroup 1 shall be selected in accordance with table XIV. In the event of a failure, the entire lot shall be inspected. Failures shall be removed from the lot. The lot shall be rejected when the number of failures in the lot is 10 percent or 1 filter connector, whichever is greater.

4.5.3.1.1 Rejected lots (subgroup 1). The contractor may rework a rejected lot to correct the defects or screen out the defective units, and resubmit the lot to the tests of subgroup 1. A lot shall be resubmitted to the tests of subgroup 1 only once. Resubmitted lots shall be separate from new lots, and shall be clearly identified as re-inspected lots. Any failures occurring in resubmitted lots consisting of fewer than 100 filter are cause for rejection of the resubmitted lot.

4.5.3.2 Subgroup 2. Samples shall be selected in accordance with table XIV. In the event of a failure, the entire lot shall be inspected. No additional failures are allowed. The failure of one contact during the contact engagement and separation force test constitutes a failure of the filter connector.

TABLE XIII. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Subgroup 1			
Thermal shock	3.20	4.6.7	Table XIV
Voltage conditioning	3.21	4.6.8	
Ground contact resistance (where applicable)	3.30	4.6.18	
Insertion loss (+25 °C)	3.19.2	4.6.9.1	
Subgroup 2			
Contact engagement and separating force (10% socket contacts only)	3.25	4.6.13	Table XIV
Coupling and uncoupling torque	3.40	4.6.28	
Crosstalk insertion loss	3.22	4.6.10	

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TABLE XIV. Sampling plan.

Lot size	Sample size
2 to 13	100%
14 to 150	13
151 to 280	20
281 to 500	29

4.6 Methods of inspection.

4.6.1 Visual and mechanical inspection. Filter connectors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.15, 3.18, 3.42, and 3.44).

4.6.2 Dielectric withstanding voltage (see 3.19.1).

4.6.2.1 Dielectric withstanding voltage (at sea level). Filter connectors shall be tested in accordance with ECIA EIA-364-20. The following details and exceptions apply:

a. Test voltage: Unless otherwise specified (see 3.1), 2.5 x rated voltage, 500 VDC.
b. Duration of application of test voltage: 1 second minimum.
c. Points of application of test voltage: Method C. The test voltage shall be applied between each contact in turn, and all other contacts and shell connected together.
d. Charging current: 5.0 milliamps (mA) maximum.
e. The sensitivity of the breakdown test equipment shall be sufficient to indicate breakdown when at least 0.5 mA of leakage current flows through the contact under test.

4.6.2.2 Dielectric withstanding voltage (at altitude). Filter connectors shall be tested in accordance with ECIA EIA-364-20, test condition IV. Unless otherwise specified (see 3.1), after 30 minutes at simulated altitude, the filter connectors shall be tested as specified in 4.6.2.1 with a test potential of 350 VDC. Only the engaging face of the connectors shall be subjected to altitude. The rear faces shall be suitably protected. Unless otherwise specified (see 3.1), the test shall be repeated at a pressure of 13.2 millimeters of mercury (90,000 feet) with a test potential of 250 VDC.

4.6.3 Insulation resistance (see 3.19.2).

4.6.3.1 Insulation resistance (+25 °C). Filter connectors shall be tested in accordance with ECIA EIA-364-21. The following details and exceptions apply:

a. Ambient temperature: +25 °C ±3 °C.
b. Test voltage: 200 VDC, applied for 2 minutes maximum.
c. Relative humidity: 20 to 50 percent.

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d. Points of measurement: Between each contact in turn, and all other contacts and shell connected together.
e. The test may be terminated once the minimum insulation resistance level has been reached. If performing with the DWV test (see 4.6.2) and the minimum insulation resistance level has been reached, the test voltage shall be applied for the minimum of one second before the test may be terminated.

4.6.3.2 Insulation resistance (+125 °C). Filter connectors shall be tested in accordance with ECIA EIA-364-21. The following details and exceptions apply:

a. Test temperature: +125 °C ±3 °C.
b. Test voltage: 200 VDC.
c. Points of measurement: Between each contact in turn, and all other contacts and shell connected together.
d. The test may be terminated once the minimum insulation resistance level has been reached. If performing with the DWV test (see 4.6.2) and the minimum insulation resistance level has been reached, the test voltage shall be applied for the minimum of one second before the test may be terminated.

4.6.4 Capacitance to ground (see 3.19.3). Capacitance shall be tested in accordance with MIL-STD-202-305. The following details and exceptions apply:

a. Test frequency: Unless otherwise specified (see 3.1), 1 kHz ±100 Hz.
b. Voltage: A root-mean-square (RMS) potential of 10-volt maximum.
c. Measurement accuracy: ±2 percent.
d. Points of measurement: Between each filtered contact in turn, and all the others connected to the filter connector shell.

4.6.5 Crosstalk capacitance (see 3.19.4). The crosstalk capacitance shall be measured in accordance with 4.6.4. The following details and exceptions shall apply:

a. Using a Kelvin 4-wire configuration, connect one set of paired leads (high) to one contact in the filter connector and the other set of paired leads (low) to an adjacent contact. Attach the ground lead to the filter connector shell.
b. Unless otherwise specified (see 3.1), no more than 3 adjacent pairs of the same capacitance value need to be measured.

4.6.6 Dissipation factor (see 3.19.5). The dissipation factor shall be measured as part of the capacitance measurement specified in 4.6.4.

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4.6.7 Thermal shock (see 3.20). Filter connectors shall be tested in accordance with ECIA EIA-364-32, method A.

a. Test condition: Test condition I, except step 3 temperature shall be +125°C +3°C, -0°C.
b. The time duration for steps 2 and 4 shall be 2 minutes maximum.
c. Number of cycles:
   (1) Qualification: 5 cycles.
   (2) Group B: 5 cycles.

4.6.8 Voltage conditioning (see 3.21). Filter connectors shall be tested in accordance with MIL-STD-202-108. The following details and exceptions apply:

a. Test temperature and tolerance: +125 °C ±3 °C.
b. Test voltage: 2 X rated voltage.
c. Test duration: 96 -0, +4 hours.
d. Mounting: Filter connector receptacles shall be mounted to a fixture (see 6.12) and mated to a plug connector or insert assembly that has been wired so that all filtered and non-filtered contacts, excluding ground contacts, are connected in parallel. The test voltage shall be applied between the paralleled contacts and the connector shell.
e. Initial measurements: Insulation resistance (+25 °C) as specified in 3.19.2.
f. With the voltage applied, the oven temperature shall be increased until the test temperature is reached. Failures shall be determined by +125 °C IR failures. The insulation resistance (+125 °C) shall be measured after the first 8 hours and then every 16 hours until the end of the test (see 6.8). This data shall be recorded for comparison purposes to determine failures. Any filter connector with an IR that has decreased by an order of magnitude or more during the test is considered a failure and marked for removal from the lot (see 4.6.8h).
g. Measurement after test: Prior to the removal of the test voltage, filter connectors shall be allowed to stabilize at room temperature (+25 °C). After stabilization at room temperature, the insulation resistance shall be measured with the contacts connected in parallel (see 4.6.8d) in accordance with 4.6.3.2. A filter connector whose IR has decreased by at least an order of magnitude or more as compared to the initial measurements (see 4.6.8e) is considered a failure and removed from the lot.
   Upon completion of the insulation resistance measurement, the connectors shall be unmated and the insulation resistance, dielectric withstanding voltage, and capacitance to ground, shall be measured in accordance with 4.6.3, 4.6.2, 4.6.4, and 4.6.6, respectively. In addition, filter connectors shall be visually inspected in accordance with 4.6.1.
   Note: If the minimum capacitance to ground requirement is not met, the filter connector may be reheated to +125 °C for a minimum of 1 hour. The capacitance to ground measurement in accordance with 4.6.4 shall be repeated after the filter connector has stabilized at room temperature (+25 °C).
   All measurements shall be conducted within 8 hours after completion of voltage conditioning (see 4.6.8).

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h. Failures during the temperature test (see 4.6.8f) may be tested for dielectric withstanding voltage and insulation resistance in accordance with 4.6.2 and 4.6.3, respectively. If the filter connector meets the requirements for DWV and IR as specified in 3.19.1 and 3.19.2, the voltage conditioning test (see 4.6.8) may be repeated. A failure during this second test is cause for lot rejection.

4.6.9 Insertion loss (see 3.19.6).

4.6.9.1 Insertion loss (+25 °C). Insertion loss shall be measured in accordance with MIL-STD-220. The following details and exceptions apply:

a. Number of contacts to be tested:
   (1) Qualification: All filtered contacts on one sample and five (filtered) on the remaining samples. For filter connectors containing multiple capacitance values (see 3.1), at least 1 contact of each capacitance value shall be tested.
   (2) Group B: Five (filtered). For filter connectors containing multiple capacitance values (see 3.1), at least 1 contact of each capacitance value shall be tested.
b. Test frequency: 100 MHz
c. Insertion loss properties shall be verified with a sweep trace in accordance with MIL-STD-220 Appendix A. Figure 2 is an example showing the performance out to 100 MHz with no indication of drop off or issues as shown on figure 3 example.

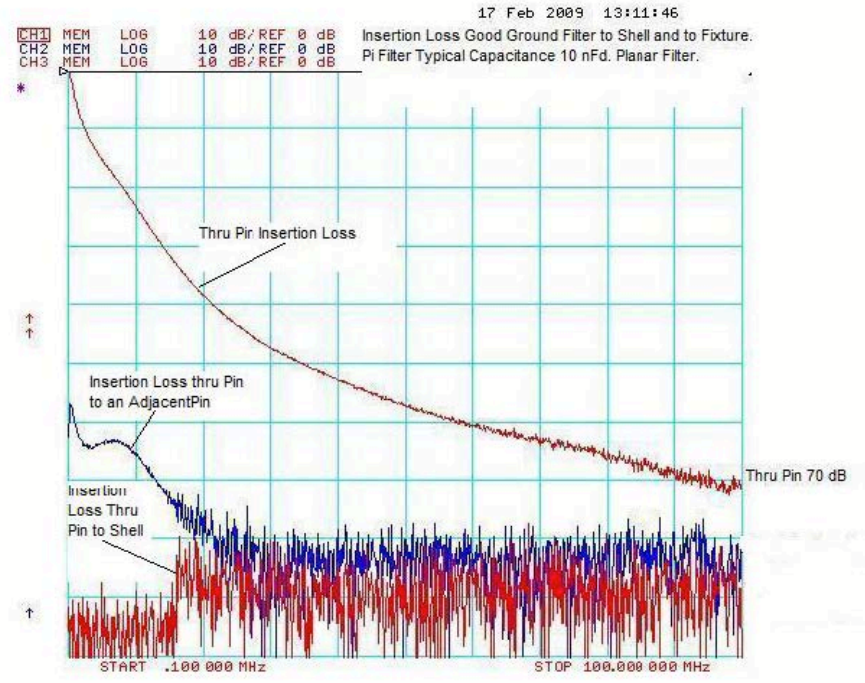


FIGURE 2. Example - Insertion loss good filter and shell ground.

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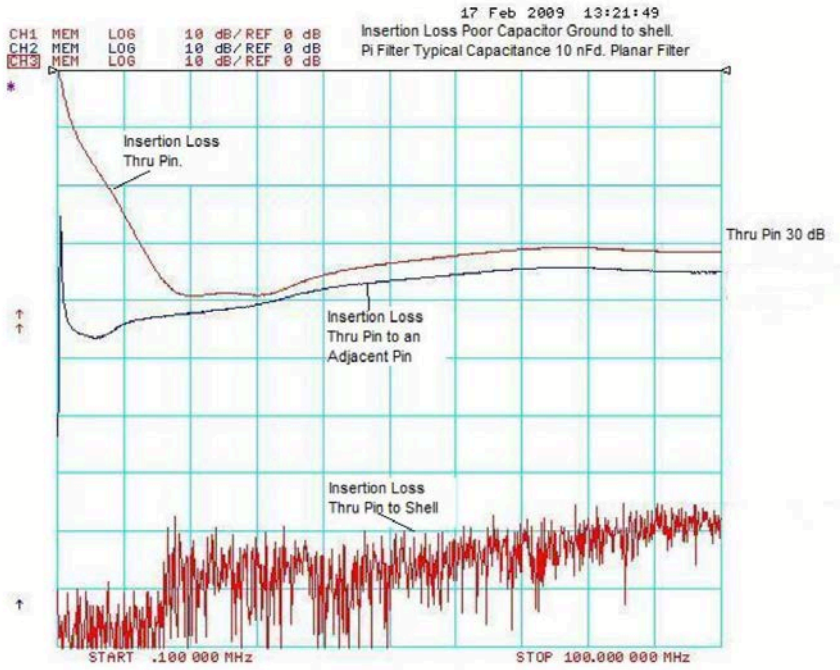


FIGURE 3. Example - Insertion loss poor filter and shell ground.

4.6.9.2 Insertion loss (+125 °C). Insertion loss shall be measured as specified in 4.6.9.1, except the filter connectors shall be exposed to +125 °C for thirty minutes prior to the test and kept at +125 °C during the test.

4.6.9.3 Insertion loss (-55 °C). Insertion loss shall be measured using the same conditions as 4.6.9.1, except the temperature shall be -55 °C. The same contacts measured at +125 °C shall be tested (see 4.6.9.2).

4.6.10 Crosstalk insertion loss (see 3.22). Crosstalk insertion loss shall be measured in accordance with MIL-STD-220. The following details and exception apply:

a. Number of contacts to be tested:
   (1) Qualification: All filtered contacts on one sample. Two separate pairs of adjacent filtered contacts on the remaining samples.
   (2) Group B: Two separate pairs of adjacent filtered contacts.
b. Test frequency: 100 MHz
c. The test signal, source output, shall be applied to a filtered contact and measured,

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source return, on an adjacent filtered contact of the same type as the filtered contact to which the test signal is applied.

d. Crosstalk performance shall be verified with a sweep trace in accordance with MIL-STD-220, Appendix A. Figure 2 is an example showing the performance out to 100 MHz with no indication of poor ground resistance or poor connection to connector shell or to filter capacitor.

4.6.11 Dielectric characteristic (see 3.23). The temperature of each filter connector shall be varied as specified in table XV. Capacitance measurements shall be made at the frequency and voltage specified in 4.6.4. The DC voltage specified in table XV shall be maintained on the filter connector during step E to step G inclusive.

TABLE XV. Voltage temperature limit cycle.

Step	Voltage (DC)	Temperature (ºC)
A	None	+25 ±2
B	None	-55 ±2
C (ref)	None	+25 ±2
D	None	+125 ±2
E	Rated	+125 ±2
F	Rated	+25 ±2
G	Rated	-55 ±2

4.6.12 Contact retention (see 3.24). Axial loads shall be applied to the filter connector contacts. The filter receptacles shall have all contacts in place during the test. A load shall be applied at a rate of approximately one pound per second until the specified load (see table XVI) has been reached. The load shall be applied to the mating ends of the contacts only. The specified loads (see table XVI) shall be maintained for a minimum of 5 seconds and then removed. Upon completion of the contact retention test, the dielectric withstanding voltage, insulation resistance, capacitance to ground, and dissipation factor shall be measured as specified in 4.6.2, 4.6.3, 4.6.4, and 4.6.6, respectively.

a. Number of contacts to be tested: A minimum of 4 filtered contacts or 20 percent of the filtered contacts, whichever is greater.
b. Measurement after test: The axial displacement of the contacts shall be measured and be in accordance with 3.24.

TABLE XVI. Contact retention.

Contact size	Load (lbs.)
22	10
20	15
16	25

4.6.13 Contact engagement and separating force (socket contacts only) (see 3.25). Filter

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connectors shall be tested in accordance with the military specification covering the connector to which it is intended to mate (see 3.1). In addition, filter connectors shall be visually inspected as specified in 4.6.1.

4.6.14 Voltage spike (see 3.26). The source impedance shall be 100 ohms. A surge generator with a 1.2 X 50 µSec voltage waveform and an 8 X 20µSec current waveform shall be used to generate the 600V spike. Upon completion of the voltage spike test, the dielectric withstanding voltage, insulation resistance, and capacitance to ground, and dissipation factor shall be measured as specified in 4.6.2, 4.6.3, 4.6.4 and 4.6.6, respectively. In addition, filter connectors shall be visually inspected as specified in 4.6.1.

4.6.15 Lightning strike (see 3.27). Filter connectors shall be tested in accordance with RTCA DO 160, section 22. The following details apply:

a. Number of contacts to be tested: Two sets of 10 filtered contacts or all filtered contacts divided into 2 sets, whichever is greater. For filtered connectors containing multiple capacitance values (see 3.1), each capacitance value shall be represented among the contacts tested.
b. Procedure: Three types of pulses (see 4.6.15c) shall be applied between the filter connector contacts under test in turn and all other contacts electrically connected to the filter connector shell. The first two pulses (see 4.6.15c (1) and 4.6.15c (2)) may be applied to the same set of filtered contacts. The third pulse (see 4.6.15c (3)) shall be applied to a separate set of filtered contacts in the same filter connector.
c. Pulses:
   (1) Exponentially decaying ringing sine at 1 MHz. A total of 10 pulses of waveform 3 level 4 shall be applied.
   (2) Double exponential pulse with positive and negative polarity per waveform 4 level 4. A total of 10 pulses in each polarity shall be applied to the contacts under test. The total will be 20 pulses.
   (3) 1600V (peak) double exponential with a short circuit current of 320 amps per waveform 2, except T2 shall be 2 µSec minimum. The pulses shall be spaced 1 second apart. The generator source impedance shall be 5 ohms. A total of 10 pulses in each polarity shall be applied to the contacts under test. The total will be 20 pulses.
d. Measurements after test: Upon completion of the lightning test, the dielectric withstanding voltage, insulation resistance, and capacitance to ground, and dissipation factor shall be measured as specified in 4.6.2, 4.6.3, 4.6.4, and 4.6.6, respectively. In addition, filter connectors shall be visually inspected as specified in 4.6.1.

4.6.16 RF current (see 3.28). A minimum of 10 percent, but not less than 3 filter contacts shall be tested individually in the filter connector as specified below:

a. Attach an RF generator through a power amplifier between the selected contact to be tested and the connector shell,

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b. Vary the RF generator frequency until RF generator shows peak current (generator loading must be chosen to provide peak current at filter resonance and also not exceed the generator voltage standing wave ratio (VSWR) tolerance). Adjust RF generator output to 0.25 amp RMS for low RF current applications, 3 amps RMS for tubular filters and 7 amps RMS for array filters and dwell for 5 minutes. There shall be no evidence of dielectric failure during or after the test.

4.6.17 Contact resistance (see 3.29). Filter connectors shall be tested in accordance with ECIA EIA-364-06. The following details and exceptions apply:

a. Number of contacts to be tested: A minimum of four filtered contacts or 20 percent of the filtered contacts, whichever is greater.
b. Test current: As specified in table VIII.
c. The contact resistance for filter connectors with Printed Circuit Board (PCB) contacts shall be measured through three inches of wire from a mating connector and a Kelvin clip attached to the appropriate PCB contact tail.

4.6.18 Ground contact resistance (see 3.30). Ground contact resistance shall be measured between each ground contact in turn, and the flange for receptacles or the end of the shell for an adapter or plug filter connector by using a pair of Kelvin clips. One clip shall be connected to the barrel of the appropriate contact, pin or socket that will mate to the contact in the mating interface of the filter connector. This will be mated to the contact to be tested and the second Kelvin clip will be connected to the connector shell at the flange or end of shell. Measurement shall be with a microhm meter with Kelvin clips or an open setup. The source will be a constant current/voltage source set at 1-volt open circuit and the current limited to 1 ampere maximum. The voltage drop shall be measured using a high impedance voltmeter and converted to resistance.

4.6.19 Water immersion (see 3.31). Filter receptacles shall be immersed unmated into water. The following details apply:

a. The water should be treated with a wetting agent such as Kodak Photo-Flow or equivalent.
b. Water temperature: +57 °C to +63 °C (+135 °F to +145 °F).
c. Duration of immersion: 20 to 40 seconds.
d. After removal from the water, the filter connector shall have free standing water removed by blowing with compressed air at 5 to 10 psi.
e. Measurements after test: Upon completion of the water immersion test, the dielectric withstanding voltage, insulation resistance, capacitance to ground, and dissipation factor shall be measured as specified in 4.6.2, 4.6.3, 4.6.4, and 4.6.6, respectively. In addition, filter connectors shall be visually inspected as specified in 4.6.1.
f. All filter connectors subjected to the water immersion test shall be vacuum dried at a maximum of 10 Torr at +82 °C (+180°F) minimum for 8 hours before resuming further testing. After vacuum baking, the dielectric withstanding voltage and insulation resistance shall be measured as specified in 4.6.2 and 4.6.3, respectively.

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4.6.20 Insert retention (see 3.32). Inserts, with grommets or any insert supporting accessories removed, shall be subjected to an axial load. Loading shall be accomplished by applying air pressure or equivalent load to the mating face. The pressure shall be increased gradually at a rate of approximately 10 psi/second until the specified pressure is reached. The insert shall retain its normal position in the shell for at least 5 seconds at the maximum specified pressure.

4.6.21 Fluid immersion (see 3.33). Unmated filter connectors and their counterpart plugs shall be fully immersed in the specified fluid. The following details apply:

a. Fluids:
   (1) Hydraulic fluid conforming to MIL-PRF-5606.
   (2) Lubricating oil conforming to MIL-PRF-7808.
b. Duration of immersion: 420 hours.
c. After removal from the fluid, each filter connector and their counterpart plugs shall remain for 1 hour in free air at room conditions.
d. Measurements after test: When mating is required, tests shall be performed on the filter connectors mated with the counterpart plugs that had been immersed simultaneously. Upon completion of the fluid immersion test, filter connectors shall be tested for coupling torque as specified in 4.6.28. Upon completion of the coupling torque test, the capacitance to ground, and dissipation factor shall be measured as specified in 4.6.4 and 4.6.6, respectively. In addition, filter connectors shall be visually inspected as specified in 4.6.1.

4.6.22 Shock (see 3.34). Filter connectors shall be tested in accordance with the military specification covering the connector to which it is intended to mate, except the filter connector shall be monitored for 1 µsec discontinuities during the test (see 6.9). Upon completion of the shock test, the dielectric withstanding voltage, insulation resistance, and capacitance to ground, and dissipation factor shall be measured as specified in 4.6.2, 4.6.3, 4.6.4 and 4.6.6, respectively. In addition, filter connectors shall be visually inspected as specified in 4.6.1.

4.6.23 Random vibration (see 3.35). Wired and mated filter connectors shall be tested in accordance with ECIA-EIA-364-28, except the filter connector shall be monitored for 1 µsec discontinuities during the test (see 6.9). The following details and exceptions apply:

a. Test condition: VI.
b. Test condition letter: J.
c. Duration and temperature: 8 hours in the longitudinal direction and 8 hours in the perpendicular direction for a total of 16 hours at +125 °C for all classes except H, Y, N hermetic receptacles. Classes H, Y, N duration shall be 4 hours in the longitudinal direction and 4 hours in the perpendicular direction for a total of 8 hours.
d. Measurements after test: Upon completion of the vibration test at each temperature, the dielectric withstanding voltage, insulation resistance, and capacitance to ground, and dissipation factor shall be measured as specified in 4.6.2, 4.6.3, 4.6.4, and 4.6.6,

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respectively. In addition, filter connectors shall be visually inspected as specified in 4.6.1.

4.6.24 Humidity (see 3.36). Filter connectors shall be tested in accordance with the military specification covering the connector to which it is intended to mate (see 3.1), except the rear of the filter connector shall be protected from the ambient of the chamber. Upon completion of the humidity test, the dielectric withstanding voltage, insulation resistance, capacitance to ground, and dissipation factor shall be measured as specified in 4.6.2, 4.6.3, 4.6.4, and 4.6.6, respectively. In addition, filter connectors shall be visually inspected as specified in 4.6.1.

4.6.25 Resistance to soldering heat (if applicable) (see 3.37). Filter connectors with solderable, non-removable contacts shall be tested in accordance with MIL-STD-202-210. The following details and exceptions apply:

Note: The resistance to soldering heat requirement may be satisfied during shock and random vibration testing (see 6.9).

a. Test condition:
   (1) Printed circuit board terminations: Test condition C.
   (2) Solder cup terminations: Test condition A.
b. Number of contacts to be tested: Seven filtered contacts or 20 percent of the filtered contacts, whichever is greater.
c. The test contacts shall be fluxed.
d. Measurements after test: Upon completion of the resistance to soldering heat test, the dielectric withstanding voltage, insulation resistance, capacitance to ground, and dissipation factor shall be measured as specified in 4.6.2, 4.6.3, 4.6.4, and 4.6.6, respectively. In addition, filter connectors shall be visually inspected as specified in 4.6.1. The visual and mechanical inspection of 4.6.1 shall be at 10X.

4.6.26 Salt spray (corrosion) (see 3.38). Filter connectors shall be tested in accordance with the military specification covering the finish of the connector to which it is intended to mate.

4.6.27 Life (see 3.39). Filter connectors shall be tested in accordance with MIL-STD-202-108. The following details and exceptions apply:

a. Test temperature and tolerance: +125 °C ±3 °C.
b. Test condition: D (1,000 hours).
c. Mounting: Filter connector receptacles shall be mounted to a fixture (see 6.12) and mated to a plug connector which has been wired so that all filtered and non-filtered contacts, excluding ground contacts, are connected in parallel. Rated voltage shall be applied between the paralleled contacts and the connector shell.
d. Initial measurements: Insulation resistance (+25 °C) as specified in 3.19.2.
e. With the voltage applied, the oven temperature shall be increased until the test temperature is reached. Failures shall be determined by +125 °C IR failures. The

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insulation resistance (+125 °C) shall be measured after the first 8 hours and then every 250 hours until the end of the test (see 6.8). This data shall be recorded for comparison purposes to determine failures. Any filter connector whose IR has decreased by an order of magnitude or more during the test is considered a failure and marked for removal from the lot (see 4.6.27g).

f. Measurement after test: Prior to the removal of the test voltage, filter connectors shall be allowed to stabilize at room temperature (+25 °C). After stabilization at room temperature, the insulation resistance shall be measured with the contacts connected in parallel (see 4.6.27c) in accordance with 4.6.3.2. A filter connector whose IR has decreased by at least an order of magnitude or more as compared to the initial measurements (see 4.6.27d) is considered a failure and removed from the lot.
Upon completion of the insulation resistance measurement, the connectors shall be unmated and the insulation resistance, dielectric withstanding voltage, capacitance to ground, and dissipation factor shall be measured in accordance with 4.6.3, 4.6.2, 4.6.4, and 4.6.6, respectively. In addition, filter connectors shall be visually inspected in accordance with 4.6.1.
Note: If the minimum capacitance to ground requirement is not met, the filter connector may be reheated to +125 °C for a minimum of one hour. The capacitance to ground measurement, in accordance with 4.6.4, shall be repeated after the filter connector has stabilized at room temperature (+25 °C).
All measurements shall be conducted within 8 hours after completion of voltage conditioning (see 4.6.8).

g. Failures during the temperature test (see 4.6.27e) may be tested for dielectric withstanding voltage and insulation resistance in accordance with 4.6.2 and 4.6.3, respectively. If the filter connector meets the requirements for DWV and IR as specified in 3.19.1 and 3.19.2, the voltage conditioning test (see 4.6.8) may be repeated. A failure during this second test is cause for rejection.

4.6.28 Coupling and uncoupling torque (see 3.40). Filter connectors shall be tested in accordance with test procedure ECIA-EIA-364-114, except filter connector and mating plug/receptacle shall be fully coupled within 5 seconds maximum, and shall be fully uncoupled within 5 seconds minimum. For qualification testing, filter connector and mating plug/receptacle shall be coupled and uncoupled. The torque applied to facilitate full coupling and uncoupling shall be measured and recorded and meet the requirements of 3.40.

4.6.29 Dielectric voltage breakdown (see 3.41). Filter connectors shall be placed in an oil (or other suitable dielectric fluid) bath and connected in series with an instrument to measure current in the mA range. A DC voltage shall be gradually applied at no greater than 50 volts per second until failure occurs. The dielectric voltage breakdown test shall be applied to not less than 3 contacts, at an external location, an internal location, and a location on the opposite side of the array. The dielectric voltage breakdown voltages shall be recorded for the samples. The mean and standard deviation shall be calculated and meet the requirements of 3.41.

4.6.30 Fungus resistance (see 3.46). Fungus testing shall be conducted in accordance with

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MIL-STD-810, Method 508 (see 6.14 to waive the fungus resistance test).

4.6.31 Insert bond strength (see 3.47). The connector's bonded insert assembly shall be tested in accordance with ECIA EIA-364-79.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Filter connectors are intended for use in the reduction of broadband radio frequency interference. Filter connectors covered by this specification are military unique because these connectors are required to operate satisfactorily in high reliability military systems that are subjected to demanding environmental conditions including high frequency vibration, high levels of shock, moisture and humidity, and temperature extremes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification, and the applicable specification sheet.
b. The complete PIN (see 1.2).
c. Packaging requirements (see 5.1).
d. Assembly instructions, if required (see 6.16).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 32633 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Naval Air Systems Command (Code 4.4.5.3), 48298 Shaw Road, Bldg. 1461, Patuxent River, MD 20670-1900 or QPL_Support@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

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6.3.1 Qualification inspection process and provisions governing qualification. To initiate the qualification process, prospective suppliers are required to forward a written request for qualification to the qualifying activity (see 6.3). The qualifying activity will respond in writing with a formal "letter of authorization" providing detailed instructions for submission of product samples. Testing cannot begin until the supplier has received this authorization letter. The qualification process, provisions, and guidelines are outlined in the SD-6, "Provisions Governing Qualification." Copies of this document are available online at https://assist.dla.mil.

6.3.2 QPL supplemental information. The qualifying activity provides a summarized list of all qualified sources on a publicly accessible electronic site. The summary includes but is not limited to the supplier approved part number and related specification part number, a dedicated approval reference number, a supplier location where purchases may be requested and the manufacturing location of the component. The suppliers and products qualified to this specification are available on the qualifying activity website at http://www.navair.navy.mil/qpl/.

6.3.3 Qualification by similarity. The supplier can submit or request a plan for qualification by similarity. This process is accomplished without completing all of the measurements, tests, and analysis requirements defined in this specification. Acceptance and the extent of similarity is determined by the qualifying activity, and is established through rationale that verifies that certain design features are identical to those being required to be tested or of a previously qualified component to the extent necessary to ensure that the component, if tested, would meet all of the required criteria. When a Qualified Products List is being established, the qualification by similarity rationale must be approved by the purchaser or qualifying activity prior to initiation of the remaining portions of the qualification process.

6.3.4 Filter designator. As part of the qualification process, the qualifying activity in the authorization letter (see SD-6) should request the manufacturer to submit the filter connector drawing defined in Appendix A. The filter designator number is assigned by the Qualifying Activity upon approval of control drawing and completion of the qualification of the filter connector. The drawing will be maintained by the qualifying activity and posted on the http://www.navair.navy.mil/qpl/ website. The drawing number consists of the MIL-DTL-32633 PIN.

6.4 Tin whisker growth. The use of alloys with a tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM B545 (Standard Specification for Electrodeposited Coatings of Tin).

6.5 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. Use of hazardous substances included in the Restriction of Hazardous Substances (RoHS) directive should be minimized.

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Included in the RoHS list are cadmium, lead, and mercury. Use of the RoHS restricted materials should be minimized or eliminated, unless needed to meet the requirements specified herein (see Section 3).

6.6 High voltage applications. For high voltage applications, the ferrites may need to have a parylene or equivalent coating to control flash over between ferrites. In some cases, the cavity containing the ferrites needs to be sealed, eliminating all voids. When a solderless design is used, the spring ends on the surface of the capacitor array need to be conformal coated to prevent edge flashover.

6.7 High vibration applications. In high vibration applications, the ferrites need to be restrained to prevent movement that can cause solder fragments to be loosened in the ferrite cavity, causing low IR readings.

6.8 Guidance for voltage conditioning and life testing (see 4.6.3, 4.6.8, and 4.6.27). Voltage conditioning and life tests both require the monitoring of insulation resistance during the time at temperature. The filter connector is to be mated to an appropriate connector or connector insert that has been wired so that all filtered and non-filtered contacts, excluding ground contacts, are connected in parallel. The test voltage is applied between the parallel contacts and the shell of the connector. The resultant insulation resistance will be the reciprocal sum of the insulation resistance of the individual contacts placed in parallel as determined by the following equation:

$$R_P = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_N}}$$

Where: N = Number of contacts in parallel.

If each individual contact had an equal insulation resistance, the parallel insulation resistance could be determined by the following equation:

$$R_P = \frac{R_C}{N}$$

Where: N = Number of contacts in parallel.
Rc = Insulation resistance of each individual contact.

For instance: If the insulation resistance of each individual contact in a 50 pin connector was 10 gigaohms, the parallel insulation resistance would be about 200 megaohms. In a practical situation, the actual level is higher because stabilization is being reached; whereas, the calculation is based on minimum limits.

The purpose of monitoring the insulation resistance during voltage conditioning and life testing is to find the contact(s) within a filter connector that have a defect that will cause the leakage current on that pin to

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increase. Since the contacts are in parallel, a circuit that has a leakage current higher than the others will cause the parallel insulation resistance measurement to change dramatically. Depending on the magnitude of the leakage, the measured insulation resistance of the paralleled pins will shift significantly out of the range of the balance of the filter connectors being monitored. For example: If the IR at temperature for the individual contacts of a 50 pin connector was 1000 megaohms, the total parallel IR would be 20 megaohms. If the IR of one pin shifted to 10 megaohms, the parallel IR would decrease to 6.7 megaohms. When degradation is taking place, it is not uncommon for the IR of an individual contact to drop 2 or even 3 decades. If more pins shifted, the 6.7 megaohms would drop even further and then begin to rise, but by small increments.

The IR limit for voltage conditioning and life testing will have to be determined by the properties of the filter connectors under test. The test voltage will be applied and then the temperature will be increased. The IR will be monitored during this time and a normal base line IR will be formed 10 to 15 minutes after the oven stabilizes. All parts should be at about the same value. Filter connectors that contain defects will suddenly shift within the next 4 to 8 hours. The shift will be significant as noted. The IR needs to be monitored after the first 8 hours every 16 hours until the end of the test.

6.9 Guidance for discontinuity monitoring during shock and random vibration testing (see 4.6.22 and 4.6.23). Discontinuity monitoring during shock and random vibration testing is accomplished by wiring the filter connector and the mating plug so that a complete series circuit is created when mated. A solder cup connector may have wires soldered to each solder cup in any sequence desired. A PCB connector may be soldered to a PCB with the traces laid out to create a series circuit when the filter connector is mated to the mating plug. The wires attached to a solder cup connector should be at least 3 feet long so that they can be secured to a non-vibrating surface adjacent to the shaker or shock table. The post electrical tests may be accomplished with the connector series wired.

A 100 milliamp maximum current flow is sent through the series circuit during vibration or shock. The 100 milliamp current is monitored for any discontinuities. A detector capable of detecting all discontinuities, which is an open circuit voltage with a duration in excess of 1 µSec, should be used.

Upon completion of the tests, the filter connectors should be unsoldered and tested per the normal test procedure. This process of soldering and unsoldering may be used to satisfy the requirements of the resistance to soldering heat test (see 3.37). Upon completion of unsoldering of the wires, care must be taken to remove any flux contamination. The flux may affect the insulation resistance if not properly cleaned.

6.10 Crimp removable contact samples. When crimp removable contacts are used, the length of stripped portion of conductor should be at least long enough to reach the bottom of the crimp barrel, but should not be so long that more than 0.031 inch (0.79 mm) of conductor is exposed at end of barrel when conductor end touches bottom of barrel. An alternate sample preparation may be used with approval of the qualifying activity.

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6.11 Capacitor characteristics (see table VII).

a. Two letter codes - MIL-PRF-123 specifies the general characteristics of ceramic dielectric fixed capacitors. Capacitor characteristics are identified by two letters as follows:

First character is a letter B - Identifies the rated temperature range.
Second character is a letter G, P, R or X - Identifies the voltage temperature limits.

A capacitor characterized as BP may be used in critical frequency applications where stability is required.

A capacitor characterized as BX may be used where variations in capacitance with respect to temperature, voltage, frequency and life can be tolerated.

b. Three letter codes - ECIA EIA RS-198 defines the standard codes for ceramic class 1 and 2 capacitors. These characteristics are defined by three letters as follows:

Class 1:

First character is a letter C, B, L, M, P, R, S, T, V or U – Identifies the change in capacitance over temperature (temperature coefficient α).

Second character is a number 0 to 8 – Identifies the multiplier of the temperature coefficient.

Third character is a letter G, H, J, K, L, M or N - Identifies the maximum tolerance of the temperature coefficient.

Class 2:

First character is a letter X, Y or Z – Identifies the low-end temperature operating temperature.

Second character is a number 4 to 9 – Identifies the high-end temperature operating temperature.

Third character is a letter P, R, S, T U or V – Identifies the capacitance change over the temperature range.

The C0G and BP ceramic capacitor characteristics are essentially the same.
The X7R and BX ceramic capacitor characteristics are essentially the same.

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6.12 Fixture (see 4.6.8 and 4.6.27). The fixture may be fabricated using two plates attached at 90 degrees with respect to each other forming a "T". The horizontal plate will have the appropriate mounting hole(s) for the receptacle connector to be tested. The exact configuration of this fixture is optional. The purpose of the fixture is to facilitate mounting of the connector for shell grounding and providing connections to the filter connector contacts. The final fixture design is at the discretion of the test activity and may be used for multiple filter connectors.

6.13 Performance modifications. The performance requirements of a filter may vary based on the application. The detail specification should include a table (see table XVII) that lists the paragraph number requirements and performance modifications to the requirement.

TABLE XVII. Example: Performance modifications.

Paragraph	Requirement	Modification	Notes
3.26	Voltage spike	Amplitude, wave form, etc. should be listed	
3.27	Lightning strike	Amplitude, waveform, etc. including ferrite coating should be listed	
3.28	RF current	Applicable current level	
3.31	Water immersion	Applicable or not and if so, modifications if any	

6.14 Fungus resistance certification. A certificate signed by an individual authorized to speak for the manufacturer may by submitted in lieu of fungus testing with the approval of the qualifying activity. The certificate must state that the filter connector components are constructed of non-nutrient materials and do not use any organic material.

6.15 Connector adapter. A connector adapter is a connector with connector interfaces on both ends. The filter connector adapter is designed to provide inline filtering in applications where the connectors already exist on the aircraft component and filtering is needed. The connector adapter interfaces between the existing aircraft equipment connector and aircraft connector.

6.15.1 Pin plug adapter. A pin plug adapter is a connector adapter designed to interface with equipment connector having a socket receptacle connector.

6.15.2 Socket plug adapter. A socket plug adapter is a connector adapter designed to interface with equipment connector having a pin receptacle connector.

6.16 Assembly instructions. If specified in the contract, the manufacturer should furnish complete assembly instructions with each connector procured under this specification. It is not the intention of this specification to require assembly instructions with un-cabled connectors (e.g., solder pot, solder tab or posts, etc.). Assembly instructions should include:

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a. Cable preparation – stripping dimensions and tolerances.
b. List and description of crimping or special tools if required.
c. Pictorial presentation of sub-assemblies and loose piece parts.
d. Sufficient pertinent dimensions for verification of correct parts; as a minimum, the cable entry openings for conductor, dielectric, braid, and jacket must be specified.
e. Recommended cable clamp tightening torque (if applicable).
f. Military PIN and manufacturer's part number.

6.17 Subject term (key word) listing.

Cadmium
Capacitor
Ceramic material
Ferrite bead
Insertion loss
Lead
Mercury

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FILTER DESIGNATOR

A.1 SCOPE

A.1.1 Scope. This appendix presents the requirements that are to be used to supplement this specification and establish the requirements for defining the filter connector designator. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 FILTER DESIGNATOR

A.2.1 Introduction. The filter designator defines the filter type and characteristic assigned to each connector contact and the ceramic material used in the design of the filter wafer assembly. As a result of different filter types, filter characteristics, and ceramic material combinations, as well as the possible high count of contacts within a filter connector, it is not feasible to define the filter designator in advance. A process for the manufacturer in coordination with the qualifying activity is required to assign a filter designator.

A.2.2 Series designator. Each connector shell size shall be assigned the first two numbers shown in table A-I followed by three numbers for the particular connector configuration which will be assigned by the qualifying activity at the time of the filter connector qualification.

A.2.2.1 Default series designator. The default series designator for each shell size shall be XX000, which shall be for a filter connector with filter type - Pi, filter characteristic – D and ceramic material characteristic – X7R. In addition, the default series designator shall have all of the contacts with the same filter type, filter characteristic, and ceramic material with no ground or feed through contacts. e.g., series designator 02000 is a size 9 shell size with all contacts being a Pi filter, D – filter characteristic and ceramic material characteristic X7R.

TABLE A-I. Series designator.

Shell size	Designator
9	02XXX
11	03XXX
13	04XXX
15	05XXX
17	06XXX
19	07XXX
21	08XXX

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A.2.3 Filter connector designator drawing. Figure A-1 is an example and provides a sample format for the filter connector designator drawing.

A.2.3.1 Filter connector insert arrangement. Figure 1 of figure A-1 is a sample pictorial of a connector insert arrangement. The insert arrangement shall be in accordance with MIL-STD-1560.

A.2.3.2 Filter connector contact identification. Table I of figure A-1 includes the contact identification, filter type, filter characteristic, ceramic material characteristic and resistance to shell assigned to each contact. This table is an example of filter contact identification and function.

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Revisions			
LTR	Description	Date	Approved

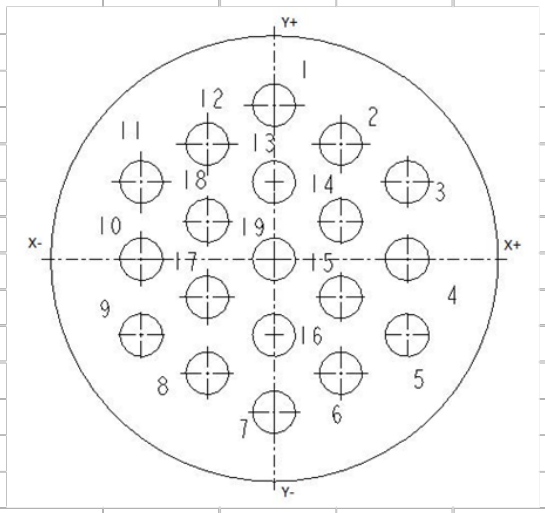


FIGURE 1. Filter connector plug insert arrangement 11-23 (see MIL-STD-1560).

	NAVAIR 4.4.5.3 PATUXTENT RIVER NAS PATUXTENT RIVER, MD 20670
	Title
PREPARED BY	
CHECKED BY	DWG NO.
APPROVED BY	MFFFFF/01W11-23PCN02000 PAGE ___ of ___

FIGURE A-1. Example - Filter connector designator control drawing.

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TABLE I. Filter contact identification and function.

Contact position ID 1/	Filter type 2/	Filter characteristic 3/	Ceramic material characteristic 4/	Resistance to shell 5/
1	Pi	D	X7R	N/A
2	Pi	D	X7R	N/A
3	Pi	D	X7R	N/A
4	Pi	D	X7R	N/A
5	Pi	D	X7R	N/A
6	Pi	D	X7R	N/A
7	Pi	D	X7R	N/A
8	Pi	D	X7R	N/A
9	Pi	D	X7R	N/A
10	Pi	D	X7R	N/A
11	Pi	D	X7R	N/A
12	Pi	D	X7R	N/A
13	Pi	D	X7R	N/A
14	Pi	D	X7R	N/A
15	Pi	D	X7R	N/A
16	Pi	D	X7R	N/A
17	Pi	D	X7R	N/A
18	Pi	D	X7R	N/A
19	Pi	D	X7R	N/A

1/ See figure 1 of figure A-1 for an example of contact position identification.
2/ See table II of figure A-1. Also see table III and figure 1 of MIL-DTL-32633 for additional information regarding filter types.
3/ See tables III, IV, and V of figure A-1.
4/ See table VI of figure A-1.
5/ Resistance to shell is the maximum voltage drop of a grounded connector contact to the connector shell. See 3.30 in MIL-DTL-32633 for additional information.

FIGURE A-1. Example - Filter connector designator control drawing - Continued.

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TABLE II. Filter type.

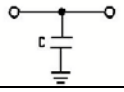
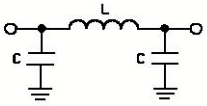
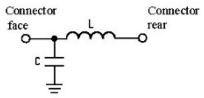
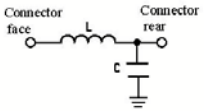
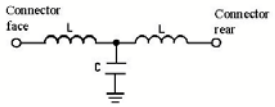
Filter type	Description
C	Single element capacitive filter. 
Pi	Three element capacitive/inductive filter, also called pi filter due to its shape that looks like the Greek letter π. 
C-L	Two element capacitive/inductive filter with the capacitor as the first element. 
L-C	Two element inductive/capacitive filter with the inductor as the first element. 
T	Three element capacitor/inductor filter, also called pi filter due to its shape that looks like the letter T. 
G	Ground – contact grounded to connector shell.
FT	Feed through – contact not filtered or grounded and is an electrical feed through the connector.

FIGURE A-1. Example - Filter connector designator control drawing – Continued.

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TABLE III. "C" type filter electrical characteristics.

Characteristic	Capacitance (pF)		Minimum insertion loss (dB)			
	Min	Max	Test frequencies			
			1 MHz	10 MHz	100 MHz	1000 MHz
A	19000	28000	6	24	41	50
B	16000	22500	5	23	39	49
C	9000	16500	3	16	35	46
D	4000	6000	-	8	28	41
E	1650	2500	-	4	21	34
F	400	650	-	-	10	23
G	200	300	-	-	5	17

TABLE IV. "Pi" type filter electrical characteristics.

Characteristic	Capacitance (pF)		Minimum insertion loss (dB)			
	Min	Max	Test frequencies			
			1 MHz	10 MHz	100 MHz	1000 MHz
H	38000	56000	10	40	62	66
I	32000	45000	8	35	60	62
J	18000	33000	5	25	57	60
K	8000	12000	1	14	50	58
L	3300	5000	-	8	40	52
M	800	1300	-	2	15	32
N	400	600	-	0.8	13	22

FIGURE A-1. Example - Filter connector designator control drawing – Continued.

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TABLE V. C-L", "L-C" and "T" type filter electrical characteristics.

Characteristic	Capacitance (pF)		Minimum insertion loss (dB)			
	Min	Max	Test frequencies			
			1 MHz	10 MHz	100 MHz	1000 MHz
O	19000	28000	6	25	42	52
P	16000	22500	6	24	41	51
Q	9000	16500	3	17	38	48
R	4000	6000	-	9	30	43
S	1650	2500	-	6	23	36
T	400	650	-	-	12	25
U	200	300	-	-	6	19

TABLE VI. Capacitance change with reference to +25°C.

Ceramic capacitor characteristic 1/	Steps A through D of table XV Bias = 0 volts	Steps E through G of table XV Bias = rated voltage
BP	0 ±30 ppm/°C	0 ±30 ppm/°C
BX	±15 percent	+15, -25 percent
C0G	0 ±30 ppm/°C	N/A
X7R	±15 percent	N/A

FIGURE A-1. Example - Filter connector designator control drawing – Continued.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
Navy - AS
Project 5935-2018-047

Review activities:
Army - AR, MI
Navy - EC, MC, OS
Air Force - 19
DLA - IS
MISC - MDA

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